1. (a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of

$$\left(2-\frac{x}{4}\right)^{10}$$

giving each term in its simplest form.

(4)

(b) Hence find the constant term in the series expansion of

$$\left(3-\frac{1}{x}\right)^{2}\left(2-\frac{x}{4}\right)^{10}$$

(3)